Prayas JEE AIR (2025)

WEEKLY TEST 21

DURATION : 180 Minutes | **DATE : 19/12/2024** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (2)
2. (1)
3. (1)
4. (3)
5. (1)
6. (3)
7. (2)
8. (2)
9. (2)
10. (1)
11. (4)
12. (4)
13. (4)
14. (4)
15. (2)
16. (3)
17. (1)
18. (2)
19. (3)
20. (1)
21. (6)
22. (18)
23. (150)
24. (10)
25. (9)

CHEMISTRY

26. (4)
27. (4)
28. (2)
29. (3)
30. (4)
31. (4)
32. (1)
33. (1)
34. (2)
35. (4)
36. (2)
37. (3)
38. (1)
39. (4)
40. (2)
41. (1)
42. (4)
43. (1)
44. (1)
45. (1)
46. (2)
47. (3)
48. (48)
49. (4)
50. (15)

MATHEMATICS

51. (1)
52. (2)
53. (4)
54. (2)
55. (2)
56. (1)
57. (2)
58. (3)
59. (2)
60. (4)
61. (4)
62. (3)
63. (4)
64. (3)
65. (2)
66. (3)
67. (3)
68. (2)
69. (4)
70. (1)
71. (21)
72. (2)
73. (7)
74. (306)
75. (3)

SECTION-I (PHYSICS)

1. **(2)**

We have,

$x = 3t$ $\qquad$ $y=5t^3$

$V_x = \frac{dx}{dt} = 3$ $\qquad$ $V_y = \frac{dy}{dt} = 15t^2$

$a_x = \frac{dV_x}{dt} = 0$ $\qquad$ $a_y = \frac{dV_y}{dt} = 30t$

$a = a_x\hat{x} + a_y\hat{y} + a_z\hat{z}$

Since, $\hat{z}$ coordinate is constant. Therefore, $a_z = 0$

$\vec{a} = 30t\hat{j}$

At $t = 1$ sec $\qquad$ $\vec{a} = 30\,\hat{j}$

2. **(1)**

Given, $\vec{a} = \frac{dv_x}{dt}\hat{x} + \frac{dv_y}{dt}\hat{y}$

$\because \frac{dv_x}{dt} = \alpha\frac{dy}{dt} = 2\alpha^2 x$

$\frac{dv_y}{dt} = 2\alpha v_x = 2\alpha^2 y$

$\vec{F} = m\vec{a} = 2m\alpha^2\left(x\hat{x} + y\hat{y}\right)$

3. **(1)**

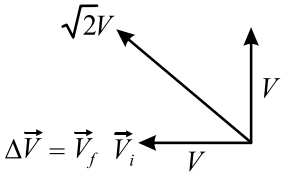
Work done = change in kinetic energy.

$-F.\,x = -K$

$x = \frac{K}{F}$

Statement I is correct

Statement II is incorrect



$\Delta\vec{V} = \vec{V}_f - \vec{V}_i$

Velocity is changing therefore, $\vec{a} \neq 0$

4. **(3)**

From 1 kg mass,

$x_{COM} = \frac{1(0) + 2.5(0) + 1.5(3)}{1 + 2.5 + 1.5} = \frac{4.5}{5} = 0.9\,\text{cm}$

$y_{COM} = \frac{1(0) + 2.5(4) + 1.5(0)}{1 + 2.5 + 1.5} = 2\,\text{cm}$

5. **(1)**

Moment of inertia of solid sphere about any tangent,

$I = I_{cm} + mr^2 \frac{2}{5}\text{mr}^2 + \text{mr}^2$

$= \frac{7}{5}\text{mr}^2$

Moment of inertia of hollow sphere about any tangent,

$I = I_{cm} + mr^2$

$= \frac{2}{3}\text{mr}^2 + \text{mr}^2 = \frac{5}{3}\text{mr}^2$

M.O.I of circular ring about its diameter, $I = \frac{MR^2}{2}$

M.O.I of circular disc about any diameter, $I = \frac{MR^2}{4}$

6. **(3)**

$F = M_{sphere}\,E_{ring} = M'\frac{Gm'\sqrt{8}R}{\left(R^2 + 8R^2\right)^{3/2}} = \frac{\sqrt{8}GMm}{27R^2}$

7. **(2)**

We know that,

Strees $\propto$ strain

$\frac{F}{A} = Y\frac{\Delta L}{L}$

$\Rightarrow \frac{F_1}{A_1} = Y_1\frac{\Delta L}{L_1}$ ….(*i*)

And $\frac{F_2}{A_2} = Y_2\frac{\Delta L}{L_2}$ …..(*ii*)

From equation (*i*) and (*ii*),

$\frac{F_1L_1}{A_2Y_1} = \frac{F_2L_2}{A_2Y_2}$

$\frac{5}{2.5\times10^{-5}\times Y_1} = \frac{6}{3\times10^{-5}\times Y_2}$

$\frac{Y_1}{Y_2} = 1$

8. **(2)**

$dVg = \rho\left(\frac{V}{2}\right)g + T(2\pi r)$

$\Rightarrow d.\frac{4}{3}\pi r^3 g = \rho.\frac{2}{3}\pi r^3 g + 2\pi rT$

$\Rightarrow \frac{2}{3}r^2 g(2d - \rho) = 2T$

$\Rightarrow r = \sqrt{\frac{3T}{(2d - \rho)g}}$

9. **(2)**

W = Area under the cyclic curve

$W = \frac{1}{2}\times 2m^3 \times 300\,Pa$

$W = 300\,Pa - m^3$

W = 300 J

10. **(1)**

Restoring force $F = -kx$

$\therefore$ A is correct

Acceleration, $a = -\omega^2 x$

B is correct

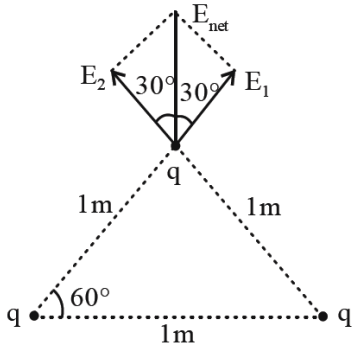
At mean position velocity is maximum

C is is correct

Acceleration is maximum at extreme points

D is incorrect.

11. **(4)**



$$E_1 = E_2 = \frac{kq}{1^2}$$

$$E_{net} = 2E_1 \cos 30°$$

$$\therefore \frac{F_{net}}{F_1} = \frac{E_{net}}{E_1} 2\cos 30° = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3} : 1$$

12. **(4)**

$$\frac{1}{2}mv^2 = mgy + \Delta U_{elec.}$$

$$\Delta U_{elec.} = kQq\left[\frac{1}{R} - \frac{1}{(R+y)}\right]$$

$$\Rightarrow v^2 = 2y\left[\frac{qQ}{4\pi\varepsilon_0 R(R+y)m} + g\right]$$

13. **(4)**

Given, $C = 1\mu F = 1 \times 10^{-6}\, F$

$V_0 = 100$ volt, $V = 50$ volt

$R = 100\ \Omega$

Where $\tau = RC = 100 \times 10^{-6}$ sec

$$50\text{V} = 100\text{V}\left(1 - e^{\frac{-t}{\tau}}\right)$$

$$\frac{1}{2} = 1 - e^{-t/\tau} \Rightarrow \frac{1}{2} = e^{-t/\tau}$$

$$2 = e^{t/\tau} \Rightarrow \ell n 2 = t/\tau$$

$$t = 0.693 \times 10^{-4} \text{ sec.}$$

14. **(4)**

Magnetic field at the centre of a semicircular arc is,

$$B_o = \frac{\mu_0 I}{4\pi R}(\pi)$$

$$= \frac{\mu_0 I}{4R} = \frac{4\pi \times 10^{-7} \times 3}{4 \times \frac{\pi}{10}}$$

$$\Rightarrow B_o = 3\mu\text{T}$$

15. **(2)**

Length of each side of the loop, $\ell = 5$ cm and, $t = 1$ sec

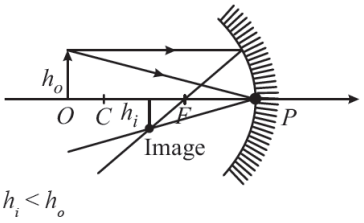
$$\therefore\ v = \frac{0.05}{1} = 0.05\,\text{m/s}$$

$$i = \frac{B\ell v}{R} = \frac{40 \times 0.05 \times 0.05}{10} = 0.01A$$

$\therefore\ F = i\ell B = 40 \times 0.01 \times 0.05 = 0.02$ N

$\therefore\ W = 0.02 \times \ell = 0.02 \times .05 = 1 \times 10^{-3}$J

16. **(3)**



$h_i < h_o$

Therefore, the image formed is real, inverted and diminished or unmagnified.

Hence, option (c) is correct answer

17. **(1)**

$$\sqrt{(2d)^2 + (d)^2} - 2d = \frac{\lambda}{2}$$

$$\Rightarrow \left(\sqrt{5} - 2\right)d = \frac{\lambda}{2} \Rightarrow d = \frac{\lambda}{2\left(\sqrt{5} - 2\right)}$$

18. **(2)**

Second and fourth statements are correct.

$\vec{E} \perp \vec{B} \perp \vec{C}$ $\quad$ $\&\left(\rho_E\right)_{eny} = \left(\rho_B\right)_{eny}$

19. **(3)**

Angular momentum, $L = \frac{nh}{2\pi}$

For Bohr orbit is, $L_1 = L = \frac{1.h}{2\pi}$ $n = 1$

In second orbit of hydrogen atom ($n = 2$),

$$L_2 = \frac{2h}{2\pi} = 2L$$

So, change momentum = = $L_2 - L_1 = 2L - L = L$

20. (1)

Let us suppose for D_4,
Atomic number = z, Mass number = A
$\therefore$ A = 182 – 4 – 4 = 174
Z = 74 – 2 – 2 + 1 = 71

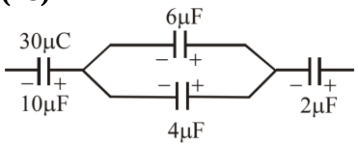
21. (6)

x_1 and x^2 are in successive loops of stationary waves.

So $\phi_1 = \pi$ and $\phi_2 = K(\Delta x)$

$$= k\left(\frac{3\pi}{2k} - \frac{\pi}{3k}\right) = \frac{7\pi}{6} = \frac{\phi_1}{\phi_2} = \frac{6}{7}$$

22. (18)



As given in the figure, 6μF and 4μF are in parallel. Now using charge conservation

Charge on 6μF capacitor $= \frac{6}{6+4} \times 30 = 18\mu C$

Since charge is asked on right plate therefore is +18μC

23. (150)

Given: R = 20Ω, f = $\frac{500}{\pi}$ Hz, ω = 1000 rad/sec.

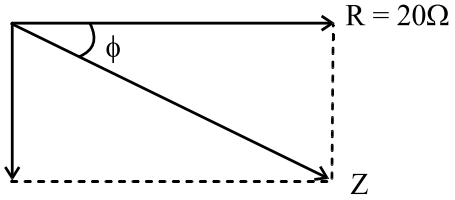
and v = 25 v

$C = 100\mu F = 100 \times 10^{-6} F$

$$x_C = \frac{1}{\omega C} = \frac{1}{1000 \times 100 \times 10^{-6}} = 10\Omega$$

$$x_L = \omega L = 1000 \times 20 \times 10^{-3} = 20\Omega$$

$$x_C - x_L = 10\Omega$$



$$z = \sqrt{R^2 + (x_C - x_L)^2} = \sqrt{400 + 100}$$

$$\cos\phi = \frac{R}{z} = \frac{20}{10\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$P_{avg} = VI\cos\phi, I = \frac{v}{z} = \frac{v^2}{z}\cos\phi = 25$$

Energy dissipated (Q) in time t = 6 sec is
Q = P.t = 25 × 6 = 150 J

24. (10)

From Einstein's photoelectric equation,

$$\frac{hc}{\lambda_1} = \phi + \frac{1}{2}m(2v)^2 \quad ...(i)$$

and $\frac{hc}{\lambda_2} = \phi + \frac{1}{2}mv^2 \quad ...(ii)$

As per question, maximum speed of photoelectrons in two cases differ by a factor 2
From eqn. (i) & (ii)

$$\Rightarrow \frac{\frac{hc}{\lambda_1} - \phi}{\frac{hc}{\lambda_2} - \phi} = 4 \Rightarrow \frac{hc}{\lambda_1} - \phi = \frac{4hc}{\lambda_2} - 4\phi$$

$$\Rightarrow \frac{4hc}{\lambda_2} - \frac{hc}{\lambda_1} = 3\phi \Rightarrow \phi = \frac{1}{3}hc\left(\frac{4}{\lambda_2} - \frac{1}{\lambda_1}\right)$$

$$3\phi = \frac{1240}{124} = 10$$

25. (9)

$\gamma_{real} = \gamma_{app.} + \gamma_{vessel}$

So $\left(\gamma_{app.} + \gamma_{vessel}\right)_{glass} = \left(\gamma_{app} + \gamma_{vessel}\right)_{steel}$

$\Rightarrow 153 \times 10^{-6} + \left(\gamma_{vessel}\right)_{glass}$

$= 144 \times 10^{-6} + \left(\gamma_{vessel}\right)_{steel}$

Further,

$\left(\gamma_{vessel}\right)_{steel} = 3\alpha = 3 \times \left(12 \times 10^{-6}\right)$

$= 36 \times 10^{-6}$ \ °C

$\Rightarrow 153 \times 10^{-6} + \left(\gamma_{vessel}\right)_{glass}$

$= 144 \times 10^{-6} + 36 \times 10^{-6}$

$\Rightarrow \left(\gamma_{vessel}\right)_{glass} = 3\alpha = 27 \times 10^{-6}$ / °C

$\Rightarrow \alpha = 9 \times 10^{-6}$ / °C

SECTION-II (CHEMISTRY)

26. (4)

Stability of hydrides decreases down the group from NH_3 to BiH_3 as M – H bond energy decreases.

27. (4)

$K_{sp} = 4(s)^3$

$= 4 \times (1.84 \times 10^{-5})^3$

$Cd(OH)_2 \rightleftharpoons Cd + 2 + 2OH^-$

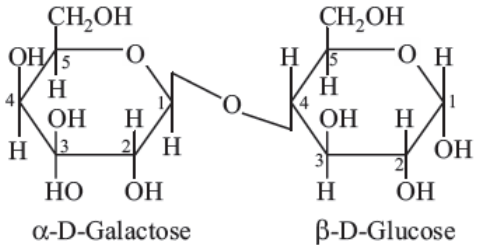
S′ S′ $(10^{-2} + S') = 10^{-2}$

$S' \times (10^{-2})^2 = 4 \times (1.84 \times 10^{-5})^3$

$S' = 4 \times (1.84)^3 \times 10^{-11}$

$(S') = 2.491 \times 10^{-10}M$

28. (2)



29. (3)

$Fe^{2+}_{(aq)} + Ag^{+}_{(aq)} \longrightarrow Fe^{3+}_{(aq)} + Ag(s)$

$E^{\circ}_{Ag^+/Ag} = xV$

$E^{\circ}_{Fe^{2+}/Fe} = yV$

$E^{\circ}_{Fe^{3+}/Fe} = zV$

Anode: $Fe^{2+}_{(aq)} \longrightarrow Fe^{3+}_{(aq)} + e^-; E^{\circ}_{Fe^{+2}/Fe^{+3}} = mV$

Cathode: $Ag^{+}_{(aq)} + e^- \longrightarrow Ag(s); E^{\circ}_{Ag^+/Ag} = xV$

Cell standard potential = (m + x)V.

$Fe^{2+} + 2e^- \rightarrow Fe;\ E^{\circ}_1 = yv \Rightarrow G^{\circ}_1 = -2Fy$

$Fe^{2+} + 3e^- \rightarrow Fe;\ E^{\circ}_2 = zv \Rightarrow \Delta G^{\circ}_2 = -3Fz$

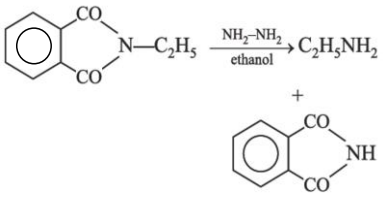
$Fe^{2+}_{(aq)} \rightarrow Fe^{3+}_{(aq)} + e^-;\ E^{\circ}_3 = mv \Rightarrow \Delta G^{\circ}_3 = -1Fma$

$\Delta G^{\circ}_3 = \Delta G^{\circ}_1 - \Delta G^{\circ}_2 = (-2Fy + 3Fz) = -Fm$

$\Rightarrow m = (2y - 3z)$

$\Rightarrow E^{\circ}_{cell} = (x + 2y - 3z)V$

30. (4)

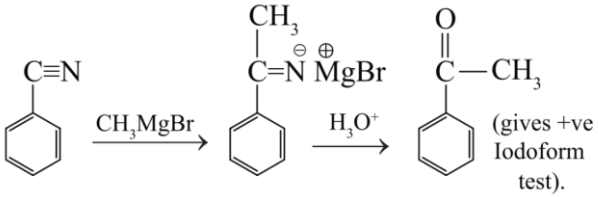


31. (4)

Ionic radii of cations is always smaller than anions. Also, more the +ve charge less will be the ionic radii, and more the –ve charge more will be the ionic radii. Hence, the correct order of ionic radii is,

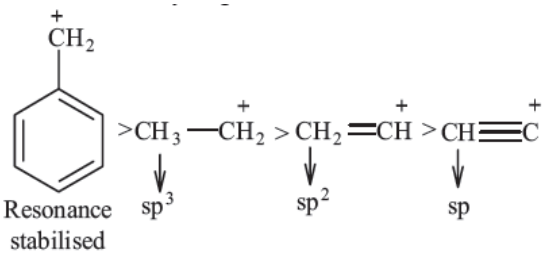
$P^{3-} > S^{2-} > Cl^- > K^+ > Ca^{2+}$

32. (1)

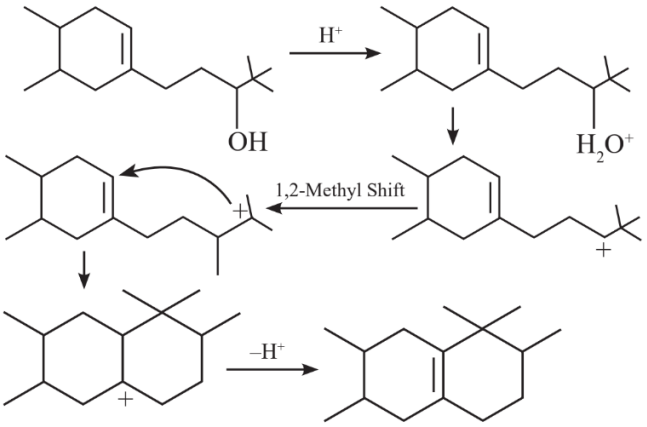


33. (1)

Order of stability of given carbocations is:



34. (2)



35. (4)

The oxidation state is the various oxoacids of 'P' are given below:

	O.S. of P
Hypophosphorous acid – H_3PO_2	+1
Orthophosphoric acid – H_3PO_4	+5
Hypophosphoric acid – $H_4P_2O_6$	+4
Orthophosphorous acid – H_3PO_3	+3

36. (2)

$C_6H_{12}O_6 \rightarrow$ Glucose

$$\% \text{ of C} = 10.8 = \frac{\text{Mass of C}}{\text{mass of solution}} \times 100$$

$$\text{Mass of C} = \frac{10.8 \times 250}{100} = 27g$$

$\therefore$ Mass of glucose = 67.5 gm

$\therefore$ Moles of glucose = 0.375 moles

Mass of solvent = 250–67.5 = 182.5 gm

$$\therefore \text{ Molality} = \frac{0.375}{0.1825} = 2.06 \text{ m}$$

37. (3)

For a zero order reaction $k = \frac{x}{t}$ ....(1)

Where x = amount decomposed

k = zero order rate constant

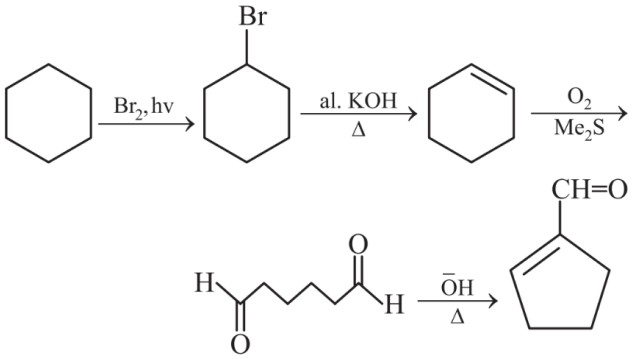
$k = \frac{[A]_0}{2t_{\frac{1}{2}}}$ ....(2)

Since $[A_0] = 2M$, $t_{1/2} = 1$ hr, k = 1

$\therefore$ from equation (1)

$t = \frac{0.25}{1} = 0.25\,hr$

38. (1)



39. (4)

Cu^{2+}, Cr^{3+} and Ni^{2+} all are paramagnetic and will coloured.

Cu^{2+} – $4s^{\circ}\ 3d^9$

Cr^{3+} – $4s^{\circ}\ 3d^3$

$Ni^{2+} = 4s^{\circ}\ 3d^8$

40. (2)

$n_{eq}.\ KMnO_4 = [FeC_2O_4 + Fe_2(C_2O_4)_3 + FeSO_4]$

or $n \times 5 = 1 \times 3 + 1 \times 6 + 1 \times 1$

$\therefore$ n = 2

41. (1)

Δ_0 is calculated from the energies of absorbed radiation not from emitted radiation (complementary colour).

42. (4)

Nucleophile, can generated by both C_2H_5OH and AgCN.

AgCN generate isonitrile as nucleophile and KCN generates, nitrile as nucleophile in nucleophilic substitution reaction.

43. (1)

For a principal quantum number n = 4, possible value of *l* = 0, 1, 2, 3. Because for any n value, *l* = 0 to n – 1, only l = 2 & l = 3 can have m = – 2. So, possible subshells are 2. Because for any value of l, m will have values from –*l* to + *l*.

44. (1)

$CoCl_3.\ 6NH_3 \longrightarrow xCl^- \xrightarrow{AlNO_3} xAgCl\downarrow$

$n(AgCl) = xn(CoCl_3.6NH_3)$

$\frac{4.78}{143.5} = x\frac{2.675}{267.5}$ $\quad \therefore x = 3$

$\therefore$ The complex is $[Co(NH_3)_6]Cl_3$

45. (1)

Vinyl group

($CH_2 = CH–$)

On ozonolysis give formaldehyde

46. (2)

On decreasing the pressure of NO by a factor of '2', the rate of reaction get decreases by a factor of '4'.

$\therefore$ Order of reaction w.r.t. 'NO' is 2.

47. (3)

According to VSEPR theory, the bent molecule is one which contains 2b.p + 1*l*.p.

N_3^- linear

NO_2^- bent (2b.q+1 l.p)

I_3^- linear

O_3 bent (2b.p + 1 l.p)

SO_2 bent

48. (48)

$MgCl_2 \rightarrow \underset{1-\alpha}{} Mg^{+2} + \underset{2\alpha}{2Cl^-}$

$i = 1 + 2\ \alpha$ ($\alpha = 0.8$)

$i = 2.6$

Acc. to relative lowering in Vapour press.

$\frac{\Delta p}{p^{\circ}} = \frac{i \times n_2}{n_1}$

$\Delta p = 2.34$

$P_s = 47.66$; $p_s \cong 48$

49. (4)

About the double bond, two geometrical isomers are possible and the compound is having one chiral carbon.

50. (15)

If $\Delta U = 0$, then,

q = – w (first law of thermodynamics)

$w = -P_{ext}\ (\Delta V) = -4.3nRT\left[\frac{1}{P_2} - \frac{1}{P_1}\right]$

$= -4.3 \times 5 \times 8.314 \times 293\left[\frac{1}{1.3} - \frac{1}{2.1}\right]$

$= -15347.70\,J/mol = -15.3\,kJ/mol$

$q = 15\,kJ/mol.$

SECTION-III (MATHEMATICS)

51. (1)

So, $x^2 - x + 27\lambda = 0$

$3x^2 - 10x + 27\lambda = 0$

α is the common root.

So, $3\alpha^2 - 10\alpha + 27\lambda = 0$ …(i)

$3\alpha^2 - 3\alpha + 6\lambda = 0$ …(ii)

Subtract $-7\alpha + 21\lambda = 0$

$-\alpha + 3\lambda = 0 \Rightarrow \alpha = 3\lambda$

By (ii) $(9\lambda^2 - 3\lambda + 2\lambda) \times 3 = 0$

$\Rightarrow 9\lambda^2 - \lambda = 0$

$\Rightarrow \lambda = 0, \frac{1}{9}$

$\lambda = \frac{1}{9}$

$x^2 - x + \frac{2}{9} = 0$

$\Rightarrow x = \frac{1}{3}, \frac{2}{3}$

and $3x^2 - 10x + 3 = 0$

$x = 3, \frac{1}{3}$

$\therefore \quad \alpha = \frac{1}{3},\ \beta = \frac{2}{3},\ \alpha = \frac{1}{3},\ \gamma = 3$

$\therefore \quad \frac{\beta\gamma}{\lambda} = \frac{\frac{2}{3}\cdot 3}{\frac{1}{9}} = 18$

52. (2)

$$\Delta = 0 = \begin{vmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}$$

$\Rightarrow \alpha^2(c-b) - \alpha(c-a) + (b-a) = 0$

It is singular when $\alpha = 1$

$$\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$$

$$= \frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a-b)(b-c)(c-a)}$$

$$= 3\frac{(a-b)(b-c)(c-a)}{(a-b)(b-c)(c-a)} = 3$$

$\left[\because a^3 + b^3 + c^3 = 3abc \text{ when } a + b + c = 0\right]$

53. (4)

$$\left(2x^2 + 3x + 4\right)^{10} = \sum_{r=0}^{20} a_r x^r$$

$$\text{General term} = \frac{10!}{r_1!r_2!r_3!}\left(2x^2\right)^{r_1}(3x)^{r_2}(4)^{r_3}$$

As a^7 = coefficient of x^7

$2r_1 + r_2 = 7$ and $r_1 + r_2 + r_3 = 10$

Possibilities are

r_1	r_2	r_3
0	7	3
1	5	4
2	3	5
3	1	6

$$a_7 = \frac{10!3^7 4^3}{7!3!} + \frac{10!(2)(3)^5(4)^4}{5!4!} + \frac{10!(2)^2(3)^3(4)^5}{2!3!5!} + \frac{10!(2)^3(3)(4)^6}{3!6!}$$

$a_{13} =$ Coeff of x^{13}

$2r_1 + r_2 = 13$ and $r_1 + r_2 + r_3 = 10$

Possibilities are

r_1	r_2	r_3
3	7	0
4	5	1
5	3	2
6	1	3

$$a_{13} = \frac{10!\left(2^3\right)\left(3^7\right)}{3!7!} + \frac{10!\left(2^4\right)\left(3^5\right)(4)}{4!5!} + \frac{10!\left(2^5\right)\left(3^3\right)\left(4^2\right)}{5!3!2!} + \frac{10!\left(2^6\right)(3)\left(4^3\right)}{6!1!3!}$$

Clearly $\frac{a_7}{a_{13}} = 2^3 = 8$

Alter Solution:-

$$\left(2x^2 + 3x + 4\right)^{10} = \sum_{r=0}^{20} a_r x^r$$

replace

$x \to \frac{2}{x}$

$$\left(2\left(\frac{4}{x^2}\right) + 3\left(\frac{2}{x}\right) + 4\right)^{10}$$

$$2^{10}\left(4 + 3x + 2x^2\right)^{10}$$

$$= \sum_{r=0}^{20} a_r \left(\frac{2}{x}\right)^r x^{20}$$

$$\left(4 + 3x + 2x^2\right)^{10}$$

$$= \sum_{v=0}^{20} a_r\, 2^{r-10} x^{20-r}$$

$a_{13} 2^3 x^7 = a_7 x^7$

$$\frac{a_7}{a_{13}} = 2^3$$

54. (2)

Let $z = x + iy$

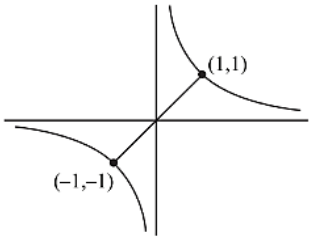
$z^4 - |z|^4 = 4iz^2$

$z^4 - |z|^4 = 4iz^2$

$\Rightarrow \quad z^4 - (z\bar{z})^2 = 4iz^2$

$\Rightarrow \quad z = 0$ or $z^2 - (\bar{z})^2 = 4i$

$\Rightarrow \quad 4ixy = 4i \Rightarrow xy = 1$

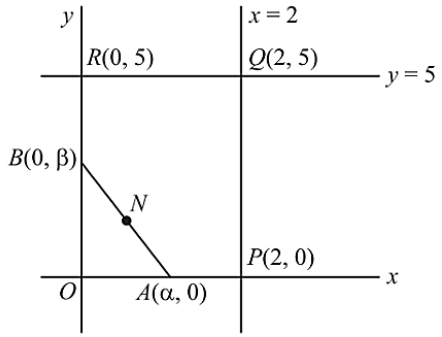


$$|z_1 - z_2|^2_{\min} = 8$$

55. (2)

$$\frac{\text{ar}(OPQR) - \text{ar}(\Delta OAB)}{\text{ar}(\Delta OAB)} = \frac{4}{1} \qquad \ldots(\text{i})$$

Let N be the mid-point of AB.



$$N(h,\ k) \equiv \left(\frac{\alpha}{2},\ \frac{\beta}{2}\right) \Rightarrow \alpha = 2h,\ \beta = 2k$$

Now from equation (i)

$$\Rightarrow \quad \frac{10 - \frac{1}{2}\alpha\beta}{\frac{1}{2}\alpha\beta} = 4$$

$$\Rightarrow \quad \frac{5}{2}\alpha\beta = 10 \Rightarrow \alpha\beta = 4$$

$$\Rightarrow \quad (2h)(2k) = 4$$

$\therefore$ Locus of N is $xy = 1$

56. (1)

Let $P = (2\lambda + 1, \lambda + 3, 2\lambda + 2)$

Let $Q = (\mu + 2, 2\mu + 2, 3\mu + 3)$

Directions ratios of PQ will be parallel to $(1, -1, -2)$

$$\Rightarrow \quad \frac{2\lambda - \mu - 1}{1} = \frac{\lambda - 2\mu + 1}{-1} = \frac{2\lambda - 3\mu - 1}{-2}$$

$$\Rightarrow \quad \lambda = \mu = 3 \Rightarrow P(7,6,8) \text{ and } Q(5,8,12)$$

$$PQ = 2\sqrt{6}$$

57. (2)

Class	Frequency	Cummulative
10-20	α	α
20-30	110	110+α
30-40	54	164+α
40-50	30	194+α
50-60	β	194+α+β
Σf=584		

Given, $\alpha + 110 + 54 + 30 + \beta = 584$

$\Rightarrow \quad \alpha + \beta = 390 \qquad \ldots(\text{i})$

$$\text{Median } = L + \left(\frac{\frac{N}{2} - C}{f}\right) \times h = 45$$

$L = 40$, $N = 584$, $C = \alpha + 164$, $f = 30$, $h = 10$

$\Rightarrow \quad \beta = 277$ From (i)

$\therefore \quad |\alpha - \beta| = 164$

58. (3)

$$a_1 + a_4 + a_7 + a_{10} + a_{13} + a_{16} = 114$$

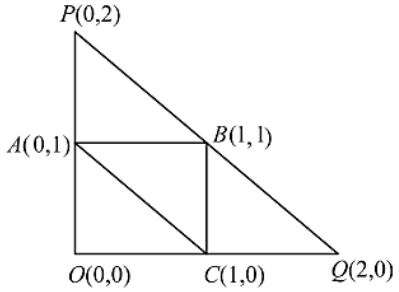
$$\frac{a_1 + a_4 + a_7 + a_{10} + a_{13} + a_{16}}{6} = \frac{a_1 + a_6 + a_{11} + a_{16}}{4}$$

$$\Rightarrow \quad \frac{114}{6} \times 4 = a_1 + a_6 + a_{11} + a_{16}$$

$$= 2 \times 38 = 76$$

59. (2)

OP = 2, OQ = 2



$$PQ = 2\sqrt{2}$$

$$D\left(\frac{4}{2+2+2\sqrt{2}}, \frac{4}{2+2+2\sqrt{2}}\right)$$

$$= D\left(\frac{2}{2+\sqrt{2}}, \frac{2}{2+\sqrt{2}}\right)$$

$\because \quad y^2 = 4ax$ passes through D

$$\therefore \quad \left(\frac{2}{2+\sqrt{2}}\right)^2 = 4a\left(\frac{2}{2+\sqrt{2}}\right)$$

$$\Rightarrow \quad 4a = \frac{2}{2+\sqrt{2}} \Rightarrow a = \frac{1}{2}\times\frac{2-\sqrt{2}}{(4-2)} = \frac{2\sqrt{2}}{4}$$

$$\text{Focus} = (a,0) = \left(\frac{2-\sqrt{2}}{4}, 0\right)$$

Also given focus $= (\alpha + \beta\sqrt{2}, 0)$

$$\therefore \quad \alpha = \frac{2}{4} = \frac{1}{2}, \beta = \frac{-1}{4}$$

Hence, $\dfrac{\alpha}{\beta^2} = \dfrac{\frac{1}{2}}{\frac{1}{16}} = 8$

60. (4)

$g(f(x))$ is differentiable at $x = 0$

$\therefore$ $g(f(x))$ must be continuous at $x = 0$.

$\Rightarrow$ $g(f(0^+)) = g(f(0^-) = g(f(0))$

$\Rightarrow$ $g(1^+) = g(-3^+) = g(1)$

$\Rightarrow$ $4 + k_2 = 9 - 3k_1$

$\Rightarrow$ $3k_1 + k_2 = 5$ …(i)

For differentiability at $x = 0$,

$g'(f(0^+))f'(0^+) = g'(f(0^-))f'(0^-)$

$\Rightarrow$ $(4)(1) = (2(-3) + k_1)(-1)$

$\Rightarrow$ $k_1=2$ and $k_2= -1$ [using(i)]

Now, $(gof)(-4) + (gof)(4) = g(f(-4) + g(f(4))$

$= g(-1) + g(e^4)$

$=(-1)^2 + 2(-1) + 4e^4 - 1$

$=4e^4 - 2 = 2(2e^4 - 1))$

61. (4)

In MATHEMATICS word, total letter =11

M comes 2 times, *A* comes 2 times, *T* comes 2 times *C* and *S* do not come together

$$= \frac{11!}{2!2!2!} - \frac{10!}{2!2!2!}\times 2!$$

$$= \frac{9\times 10!}{2!2!2!} = \frac{9\times 10\times 9\times 8\times 7\times \underline{|6}}{8}$$

$$= 5670\underline{|6} = k\underline{|6}$$

Hence, k is 5670

62. (3)

$$\cot^{-1}\alpha = \cot^{-1}2 + \cot^{-1}8 + \ldots \text{ up to } 100 \text{ terms}$$

$$= \sum \cot^{-1} 2n^2 = \tan^{-1}\frac{1}{2n^2}$$

$$\sum_{n=1}^{100} \tan^{-1}\left[\left(\tan^{-1}3 - \tan^{-1}1\right) + \left(\tan^{-1}5 - \tan^{-1}3\right)\right.$$

$$= \tan^{-1}201 - \tan^{-1}1$$

$$\tan^{-1}\frac{200}{202} = \cot^{-1}\alpha = \cot^{-1}\frac{101}{100} = \alpha = 1.01$$

63. (4)

Given $R = \{(a, b), (b, c), (b, d)\}$

In order to make it equivalence relation as per given set, R must be

$\{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (b, c), (c, b), (b, d), (d, b), (a, c), (a, d), (c, d), (d, c), (c, a), (d, a)\}$)

Three already given so 13 more to be added.

64. (3)

$$(x_1, y_1) = \left(\frac{a\sum\cos\alpha}{3}, \frac{b\sum\cos\beta}{3}\right) = \text{circumcenter}$$

$$\sum\cos\alpha = \frac{3x_1}{a} \qquad \ldots(1)$$

$$\sum\cos\alpha = \frac{3y_1}{b} \qquad \ldots(2)$$

Square and add

Hence, the correct answer is (3).

65. (2)

$\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$

$b = \hat{i} + 3\hat{j} + 5\hat{k}$

Projection of $\vec{a}$ on $\vec{b}$ is

$$\vec{a}\cdot\hat{b} = \frac{5-3-15}{\sqrt{35}} = \frac{-13}{\sqrt{35}}$$

66. (3)

Total number of possibilities $n(S) = 2^5 \times 2^5$

Favourable case $n(E) = {}^5C_2 \times 3^3 = 10 \times 3^3$

$\therefore$ required probability

$$P(E) = \frac{10\times 3^3}{2^5\times 2^5} = \frac{5\times 27}{2^9} = \frac{135}{2^9}$$

67. (3)

$$2x = 4by' \Rightarrow b = \frac{x}{2y'}$$

So, differential equation is $x^2 = \dfrac{2x}{y'}y + \left(\dfrac{x}{y'}\right)^2$

$$\Rightarrow x\left(\frac{dy}{dx}\right)^2 = 2y\frac{dy}{dx} + x \text{ or } x\left(y'\right)^2 = 2yy' + x$$

68. (2)

Given, $y - x = 2$ and $x^2 = y$

$\Rightarrow$ $x^2 = x + 2$

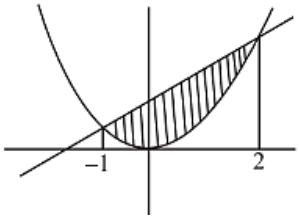
$\Rightarrow$ $x^2 - x - 2 = 0$

$\Rightarrow$ $(x+1)(x-2) = 0$

$\Rightarrow$ $x = -1, 2$

Required area $= \int_{-1}^{2}\left[(x+2)-x^2\right]dx$

$= \left[\frac{x^2}{2}+2x-\frac{x^3}{3}\right]_{-1}^{2} = 6-\frac{8}{3}-\frac{1}{2}+2-\frac{1}{3}=\frac{9}{2}$



69. (4)

Given, $f(x)=\begin{cases}\left(2-\sin\frac{1}{x}\right)|x|, & x\neq 0\\ 0, & x=0\end{cases}$

gathered $f(x)=\begin{cases}x\left(\sin\frac{1}{x}-2\right), & x<0\\ 0, & x=0\\ x\left(2-\sin\frac{1}{x}\right), & x>0\end{cases}$

$f'(x)=\begin{cases}-2+\sin\frac{1}{x}-\frac{1}{x}\cos\frac{1}{x}, & x<0\\ 2-\sin\frac{1}{x}+\frac{1}{x}\cos\frac{1}{x}, & x>0\end{cases}$

$f'(0)$ is undefined.

70. (1)

$(2021)^{2022} + (2022)^{2021} = (2023–2)^{2022} + (2023–1)^{2021}$

$=7k_1+2^{2022} + (2023–1)^{2021}$

$=7k_1 + 2^{2022} + 7k_2–1$

$=7(k_1 + k_2) + 8^{674}–1$

$=7(k_1 + k_2) + 7k_2 + 1–1$

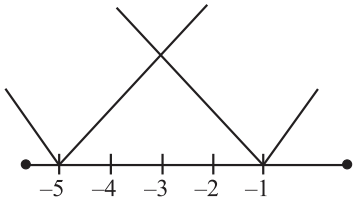
$=7(k_1 + k_2 + k_3))$

So, the given number is divisible by 7.

Hence, the remainder is zero.

71. (21)

$\int_{-6}^{0} f(x)dx = \int_{-6}^{-3}|x+1|\,dx + \int_{-3}^{0}|x+5|\,dx$



$= -\int_{-6}^{-3}(x+1)dx + \int_{-3}^{0}(x+5)dx$

$= -\left[\frac{x^2}{2}+x\right]_{-6}^{-3} + \left[\frac{x^2}{2}+5x\right]_{-3}^{0}$

$= -\left[\left(\frac{9}{2}-3\right)-(18-6)\right] + \left[0-\left(\frac{9}{2}-15\right)\right]$

$= -\left[\frac{3}{2}-12\right] + \frac{21}{2} = \frac{21}{2}+\frac{21}{2} = 21$

72. (2)

$C_1 : x^2 + y^2 - 4x - 2y + (5-\alpha) = 0$

Centre $O_1 = (2,1)$, radius$(r_1) = \sqrt{\alpha}$

$C_2 : x^2 + y^2 - 2fx - 2gy + \frac{36}{5} = 0$

Centre $O_2 = (f, g)$ radius $(r_2) = \sqrt{f^2+g^2-\frac{36}{5}}$

$\Rightarrow \frac{f-2}{2} = \frac{g-1}{-1} = -2\cdot\left(\frac{2\times 2-1+1}{2^2+1^2}\right) = \frac{-8}{5}$

$\Rightarrow f = \frac{-6}{5}$ and $g = \frac{13}{5}$

$\Rightarrow \alpha = 1$ and $r = 1$

Hence, $\alpha = r = 2$.

73. (7)

Let $I = \int\frac{2e^x+3e^{-x}}{4e^x+7e^{-x}}dx = \int\frac{2e^{2x}+3}{4e^{2x}+7}dx$

$= \int\frac{2e^{2x}}{4e^{2x}+7}dx + \int\frac{3e^{-2x}}{4+7e^{-2x}}dx$

Put $4e^{2x}+7 = t \Rightarrow 8e^{2x}dx = dt$

$\Rightarrow 4+7e^{-2x} = \lambda \Rightarrow -14e^{-2x}dx = d\lambda$

$= \frac{1}{4}\int\frac{dt}{t} - \frac{3}{14}\int\frac{d\lambda}{\lambda} = \frac{1}{4}\ln t - \frac{3}{14}\ln\lambda + c$

$= \frac{1}{4}\ln\left(4e^{2x}+7\right) - \frac{3}{14}\ln\left(4+7e^{-2x}\right) + c$

$= \frac{1}{14}\left[\frac{7}{2}\ln\left(4e^{2x}+7\right) - 3\ln\left(4+7e^{-2x}\right)\right] + c$

$= \frac{1}{14}\left[\frac{7}{2}\ln\left(e^x\left(4e^x+7e^{-x}\right)\right) - 3\ln\left\{e^{-x}\left(4e^x+7e^{-x}\right)\right\}\right] + c$

$= \frac{1}{14}\left[\frac{1}{2}\ln\left(4e^x+7e^{-x}\right) + \frac{13}{2}x\right] + c$

Also, it is equal to

$\frac{1}{14}\left[ux + v\log_e\left(4e^x+7e^{-x}\right)\right] + c$

On comparing we get

$\Rightarrow u = \frac{13}{2}, v = \frac{1}{2}$

Hence, $u + v = 7$

74. **(306)**

$$H_n = \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1$$

$$e = \sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+\frac{3+n}{1+n}} = \sqrt{\frac{2n+4}{n+1}}$$

$$e = \sqrt{\frac{2n+4}{n+1}}$$

n =48 (smallest even value for which $e \in Q$)

$$\Rightarrow \; e = \frac{10}{7}$$

Now compare H_n with standard equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$a^2 = n+1, b^2 = n+3$$

$$a^2 = 49; b^2 = 51$$

$$l = \text{ length of } LR = \frac{2b^2}{a}$$

$$\Rightarrow \; l = 2 \cdot \frac{51}{7}$$

$$\Rightarrow \; l = \frac{102}{7}$$

$$\Rightarrow \; 21l = 306$$

75. **(3)**

$$96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{2^2\pi}{33}\cos\frac{2^3\pi}{33}\cos\frac{2^4\pi}{33}$$

$\because \quad \cos A \cos 2A \cos 2^2 A \cos 2^3 A \ldots \cos 2^{n-1}$

$$A = \frac{\sin\left(2^n A\right)}{2^n \sin A}$$

Here, $A = \frac{\pi}{33}, n = 5$

$$\frac{96\sin\left(2^5 \frac{\pi}{33}\right)}{2^5 \sin\left(\frac{\pi}{33}\right)} = \frac{96\sin\left(\frac{32\pi}{33}\right)}{32\sin\left(\frac{\pi}{33}\right)}$$

$$= \frac{3\sin\left(\pi - \frac{\pi}{33}\right)}{\sin\left(\frac{\pi}{33}\right)} = 3$$



PW Web/App - https://smart.link/7wwosivoicgd4

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